

Reproduction of ultralytics/ultralytics: YOLO11n fine-tuning on COCO128

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Submitted by K-Veritas Team

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Independent reproduction · Benchmark or leaderboard entry · Computer vision

Abstract

Fine-tuned the pretrained YOLO11n detector on COCO128 with the yolo CLI from the pinned commit (installed editable): 100 epochs, image size 640, default hyperparameters, seed 0, plots disabled; then validated the best checkpoint with yolo detect val. Per-epoch box loss and mAP from results.csv; final precision, recall, mAP50 and mAP50-95 from the validation run. COCO128 and yolo11n.pt fetched by Ultralytics before the sealed run. Single NVIDIA RTX 5060 Ti 16 GB, PyTorch 2.11 cu128, Python 3.12.

About this record

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1 Sealed reports

The experiments were sealed with K-Veritas [1], which binds each result to the code, hardware and time that produced it.

Report	Label	Sealed	Runs	Invocations	Signer
r1 [2]	YOLO11n COCO128	2026-10-07	1	1	Signed by the K-Veritas server

2 Results

r1: YOLO11n COCO128

Run	Metric	Final value	Step
1	train_box_loss	0.82523	100
1	mAP50	0.89164	100
1	mAP50_95	0.7281	100

Run	Inline claim	Value	Phase
1	val_precision	0.916	
1	val_recall	0.821	
1	val_mAP50	0.896	
1	val_mAP50_95	0.746	

3 Setup

Report	Item	Value
r1	Operating system	linux / amd64
r1	CPU	Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz, 16 cores
r1	GPU	NVIDIA GeForce RTX 5060 Ti, 16311 MiB
r1	Memory	31.0 GB
r1	Total run time	190.1 s

4 Files

Each report and code bundle must hash to these values. A bundle belongs to the report whose signed data names its hash.

File	SHA-256	Size
r1.pdf	6962f66e63387bb94c038fd38145780824a9b37d8ef03b86c056d36dba8d0736	2.3 MB
r1.kvbundle.zip	bdd0afbbaa4d1aa71a9873dfe1774952f50e90b3ca466684e4ed98807d2eb7e5c	1.3 MB

5 Links and licence

Reproduction of: <https://github.com/ultralytics/ultralytics/tree/ce1399439b93c490b648b55dedd0a6334f0199da>

Record: CC-BY 4.0. Code: AGPL-3.0, as declared by the authors. Tags: yolo, object-detection, coco128.

Listed, not endorsed. Review checks the artifact, never the science. Reports are never edited; changing them makes a new version. Other fields are corrected in place, and every correction is logged in metadata.yaml.

References

[1] M. K. Keita and C. Homan. Computer Science Conferences Should Require Nonrepudiable Experimental Results. NeurIPS 2026 Position Paper Track. arXiv:2605.08586.

[2] Sealed experiment report r1 (YOLO11n COCO128), kv:2610.00008v1/r1. Data hash

65d73c52426217eb78e207d2e9ac92248ce112fe912cc8edf6e75030ee2edc13.

<https://kveritas.org/records/2610.00008v1/r1.pdf>